Review plan · for the teacher

Leading data science at school: unit check

Data science · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Samples and margins
- Finding groups
- Fair models
- Deciding with data

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea will you bring to staff first?
- Which school tool will you audit?
- Which hunch will you test?

Exit ticket

- One trial I will propose this term, and its comparison group.

Try it at school tomorrow
Bring one trial plan to your next staff meeting and agree on a date to share its results.

Quiz sheet · for each team

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Team _____ Date _____

1 **Two fair random samples always give the same average.**

◯ True ◯ False

2 **A sample is four times bigger. What happens to the margin of error?**

◯ It quarters ◯ It halves ◯ It doubles

3 **Put the k-means steps in order.**

☐ Join the nearest centre

☐ Place k centres

☐ Repeat

☐ Move centres to the mean

4 **A survey only asked pupils at the after-school club. What is the main problem?**

◯ It is too small ◯ It used paper ◯ It is not random, so some pupils are missing

5 **A cluster's name, such as "keen readers", comes from us, not from the computer.**

◯ True ◯ False

6 **Work out the error rates.**

Group A: 2 wrong in 20 is _____ %. Group B: 8 wrong in 20 is _____ %.

Word bank: 2 · 40 · 10 · 8

7 **Which help find or reduce bias? Choose all that apply.**

☐ More varied training data ☐ A person checks important results ☐ Reporting error rates by group

☐ Only reporting overall accuracy

8 **Match each term to its meaning.**

Comparison group	• • • •	How far a sample may be from the truth
Effect	• • • •	Similar cases kept the same
Margin of error	• • • •	Trial change minus comparison change

9 **Work out the effect.**

Trial: 20 → 14. Comparison: 19 → 17. The effect is _____ fewer.

Word bank: 6 · 4

10 **Who should make the final school decision?**

◯ The supplier of the model ◯ People, weighing data and voices ◯ The data alone

Answers · for the teacher

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1 **False**

Samples wobble by chance.

2 **It halves**

The margin shrinks with the square root of the sample size.

3 **Top to bottom: 2, 1, 4, 3**

Assign, move, repeat.

4 **It is not random, so some pupils are missing**

Who is asked matters more than how many.

5 **True**

The computer finds similar numbers; the names are our interpretation.

6 **Group A: 2 wrong in 20 is 10%. Group B: 8 wrong in 20 is 40%.**

2 ÷ 20 = 10% and 8 ÷ 20 = 40%: the model is unfair to group B.

7 **More varied training data; A person checks important results; Reporting error rates by group**

Varied data reduces bias, a person catches biased results, and error rates by group reveal it; overall accuracy alone hides it.

8 **Comparison group — Similar cases kept the same**
Effect — Trial change minus comparison change
Margin of error — How far a sample may be from the truth

Key ideas for fair evidence.

9 **Trial: 20 → 14. Comparison: 19 → 17. The effect is 4 fewer.**

(20 − 14) − (19 − 17) = 6 − 2 = 4.

10 **People, weighing data and voices**

Data informs; people decide.